

SQL Project

Tech
Instagram
Influencer
Account
Analysis.

Dataset Overview

For this project, the analysis was performed on an Instagram Influencer Activity dataset designed to evaluate account performance, audience engagement, and content effectiveness. The dataset contains detailed information on Instagram account activity, follower growth, profile engagement, and post-level performance metrics. The primary goal of the dataset is to help understand how different types of content perform and how audience interaction contributes to influencer growth over time. The database is structured using **three relational tables**, which are interconnected through the date column and together provide a complete view of the influencer account's activity and engagement patterns. The **three tables** included in the dataset are:

☐ **Dim date Table**

The dim date table serves as a **date dimension table, containing** calendar-related information to support time-based analysis.

☐ **Fact account Table**

The fact account table contains high-level Instagram account performance metrics. This table focuses on the overall growth and visibility of the influencer account rather than individual posts. Indicating strong audience engagement during that period.

☐ **Fact content Table**

The fact content table is the most detailed and important table in the dataset. It contains post-level performance data and engagement metrics for different types of Instagram content.

TASK-1

How many unique post types are found in the fact content table?

RESULT OF QUERY 1

	post_type
▶	IG Image
	IG Reel
	IG Carousel
	IG Video

INTERPRETATION OF QUERY 1

- ❑ There are four different types of Instagram posts in this dataset: **IG Image**, **IG Reel**, **IG Carousel**, and **IG Video**.

TASK-2

What are the highest and lowest recorded impressions for each post type?

RESULT OF QUERY 2

	post_type	Highest_Impression	Lowest_Impression
▶	IG Carousel	9677	3264
	IG Image	129694	23367
	IG Reel	339708	87570
	IG Video	73321	8741

INTERPRETATION OF QUERY 2

- ❑ The data clearly indicates that **IG Reels are the most effective content format for maximizing impressions and audience reach**. If the objective is to increase visibility and attract a larger audience, focusing on Reel-based content would be the most effective strategy. Traditional image posts can serve as a strong secondary content type, while videos and carousels may require optimization to improve their performance.

TASK-3

Filter all the posts that were published on a weekend in the month of March and April and export them to a separate csv file.

RESULT OF QUERY 3

date	post_category	post_type	video_duration	carousel_item_count	impressions	reach	shares	follows	likes	comments	saves	date	month_name	weekday_name	weekday_or_weekend	week_no
2023-03-04	Earphone	IG Video	291	0	12265	3668	69	92	327	7	18	2023-03-04	March	Saturday	Weekend	W9
2023-03-05	Smartwatch	IG Image	0	0	62770	18001	273	360	1194	28	76	2023-03-05	March	Sunday	Weekend	W10
2023-03-11	Mobile	IG Carousel	0	3	5899	1093	45	12	53	0	6	2023-03-11	March	Saturday	Weekend	W10
2023-03-12	Laptop	IG Image	0	0	79416	23474	327	259	1235	69	204	2023-03-12	March	Sunday	Weekend	W11
2023-03-18	Mobile	IG Carousel	0	3	9157	2254	67	58	55	6	15	2023-03-18	March	Saturday	Weekend	W11
2023-03-19	Smartwatch	IG Carousel	0	3	4146	1079	42	17	43	1	6	2023-03-19	March	Sunday	Weekend	W12
2023-03-25	Earphone	IG Reel	22	0	132284	66721	1093	1482	3622	83	695	2023-03-25	March	Saturday	Weekend	W12
2023-03-26	Mobile	IG Image	0	0	63425	26113	435	336	1994	68	179	2023-03-26	March	Sunday	Weekend	W13
2023-04-01	Mobile	IG Carousel	0	3	4549	1052	27	18	35	1	6	2023-04-01	April	Saturday	Weekend	W13

INTERPRETATION OF QUERY 3

- ❑ Content format significantly influences Instagram performance. IG Reels are the top-performing format, driving the highest reach, impressions, engagement, and follower growth. IG Images also perform well, generating strong audience interactions through likes and comments. Meanwhile, IG Carousels and Videos contribute to content variety but deliver comparatively lower visibility. To maximize growth and engagement, brands should prioritize Reels, supported by Images for consistent audience interaction.

TASK-4

Create a report to get the statistics for the account. The final output includes the following fields:

- Month Name • Total Profile Visits • Total New Followers

RESULT OF QUERY 4

	Month_Name	Profile_Visit	New_Followers
►	January	26512	17053
	February	20628	15254
	March	23132	18285
	April	29852	21799
	May	106571	66984
	June	103350	76942
	July	54352	33302
	August	42094	24371
	September	41522	28523

INTERPRETATION OF QUERY 4

- ☐ May and June were the strongest growth months, with May achieving the highest profile visits (106,571) and June recording the highest new followers (76,942). Performance remained steady from January to April, while profile visits and follower growth declined after June but stayed above early-year levels.

TASK-5

Write a CTE that calculates the total number of 'likes' for each post category' during the month of 'July' and subsequently, arrange the post category' values in descending order according to their total likes.

RESULT OF QUERY 5

	post_category	Total_Likes
▶	Other Gadgets	26519
	Tech Tips	20296
	Mobile	16338
	Earphone	14435
	Smartwatch	3918

INTERPRETATION OF QUERY 5

- ❑ Among the post categories, **Other Gadgets and Tech Tips received the highest total likes**, indicating stronger audience engagement and interest compared to Mobile, Earphone, and Smartwatch content. This suggests that informative and diverse technology-related content resonates most with the audience.

TASK-6

Create a report that displays the unique post category names alongside their respective counts for each month. The output should have three columns

- month name
- post category names
- post category count.

RESULT OF QUERY 6

	Month_Name	Post_Category_Names	Post_Category_Count
►	May	Earphone,Laptop,Mobile,Other Gadgets,Smartwatch,Tech Tips	6
	April	Earphone,Laptop,Mobile,Other Gadgets,Smartwatch	5
	August	Earphone,Mobile,Other Gadgets,Smartwatch,Tech Tips	5
	July	Earphone,Mobile,Other Gadgets,Smartwatch,Tech Tips	5
	February	Earphone,Laptop,Mobile,Smartwatch	4
	June	Mobile,Other Gadgets,Smartwatch,Tech Tips	4
	March	Earphone,Laptop,Mobile,Smartwatch	4
	September	Mobile,Other Gadgets,Smartwatch,Tech Tips	4
	January	Earphone,Mobile,Smartwatch	3

INTERPRETATION OF QUERY 6

- ❑ Content diversity increased significantly from the start of the year, peaking in May with all six categories represented. Mobile and Smartwatch remained core content pillars throughout the year, while category focus evolved based on audience interests and content performance.

TASK-7

What is the percentage breakdown of total reach by post type? The final output includes the following fields:

- post type • total reach • reach percentage

RESULT OF QUERY 7

	post_type	Total_Reach	Reach_Percentage
▶	IG Reel	5379091	61.63
	IG Image	1866381	21.38
	IG Video	1422300	16.30
	IG Carousel	60465	0.69

INTERPRETATION OF QUERY 7

- ❑ **IG Reels and IG Image posts** generated significantly **higher reach** compared to Videos and Carousel posts, indicating that these content formats are more effective at reaching a larger audience. This suggests that prioritizing Reels and Image-based content could help maximize visibility and audience engagement.

TASK-8

Create a report that includes the quarter, total comments, and total saves recorded for each post category. Assign the following quarter groupings:

(January, February, March) → "Q1"

(April, May, June) → "Q2"

(July, August, September) → "Q3".

The final output columns should consist of:

- post category • quarter • total comments
- total saves

RESULT OF QUERY 8

	post_category	Quart	Total_Comments	Total_Saves
▶	Earphone	Q1	351	2230
	Earphone	Q2	589	3602
	Earphone	Q3	427	3247
	Laptop	Q1	418	2837
	Laptop	Q2	452	2248
	Mobile	Q1	1836	9843
	Mobile	Q2	2313	17207
	Mobile	Q3	1134	5285
	Other Gadgets	Q2	1622	12041

INTERPRETATION OF QUERY 8

- ❑ Mobile was the highest-performing category, achieving the most comments and saves, with peak engagement in Q2. Other Gadgets also performed strongly in Q2, particularly in saves, highlighting high content value. Earphone content grew from Q1 to Q2 before declining slightly in Q3, while Laptop posts maintained steady but moderate engagement. Overall, **Q2 was the strongest quarter across categories**, generating the highest audience interaction and content saves.

TASK-9

List the top three dates in each month with the highest number of new followers. The final output should include the following columns:

• month • date • new followers

RESULT OF QUERY 9

	Month_Name	date	New_Followers
▶	January	2023-01-30	3186
	January	2023-01-03	2959
	January	2023-01-23	1003
	February	2023-02-01	4106
	February	2023-02-24	2383
	February	2023-02-02	1989
	March	2023-03-21	5421

INTERPRETATION OF QUERY 9

- ❑ May and June generated the highest follower growth, with June contributing the most top-performing days, indicating strong audience engagement and effective content performance during this period.

TASK-10

Create a stored procedure that takes the Week no as input and generates a report displaying the total shares for each Post type. The output of the procedure should consist of two columns:

- Post type
- Total shares

RESULT OF QUERY 10

	post_type	total_shares
▶	IG Image	861
	IG Video	410
	IG Carousel	42

This result only shows 'W1'.

INTERPRETATION OF QUERY 10

☐ **Week 1 Insight:**

In Week 1, **IG Image** was the top-performing post type, generating **861 total shares**. This indicates that image-based content received the highest share engagement during the first week.

- ☐ If we call the stored procedure again for W2, W3, and so on, then we get the results for other weeks..

Key Insights and Strategy

- ❑ **IG Reels were the best-performing format**, driving the highest reach, impressions, engagement, and follower growth.
- ❑ **IG Images ranked second**, generating strong likes and comments, making them effective for audience interaction.
- ❑ **IG Reels and Images significantly outperformed Carousels and Videos** in terms of visibility and reach.
- ❑ **May and June were the strongest growth months**, with May recording the highest profile visits and June achieving the highest follower gain.
- ❑ **Other Gadgets and Tech Tips received the most likes**, indicating strong audience interest in informative technology content.
- ❑ **Mobile content generated the highest comments and saves**, making it the most engaging content category.
- ❑ **Q2 was the strongest-performing quarter**, delivering the highest engagement, saves, comments, and follower growth.
- ❑ **Content diversity peaked in May**, while Mobile and Smartwatch remained consistent content pillars throughout the year.
- ❑ **June contributed the highest number of top-performing days**, highlighting peak audience engagement and content effectiveness.
- ❑ Focus on **IG Reels and Images**, prioritize **Mobile, Other Gadgets, and Tech Tips** content, and replicate the successful strategies used during **May and June** to maximize growth and engagement.

*THANK
YOU ...*

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